7.2.2 Key architectural components of RAG

Figure 7.8 illustrates the general architecture of a typical RAG system that surfaces knowledge from a list of documents and allows users to query information using a chat-like environment.

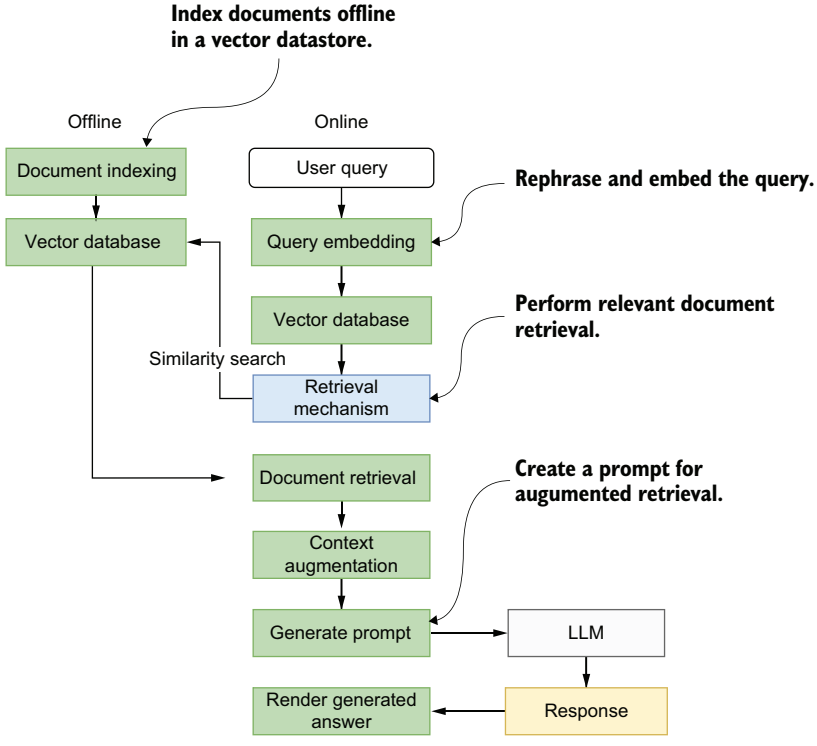


Figure 7.8 Architecture of a RAG system. The system first indexes a collection of documents offline into a vector database, enabling efficient semantic search. When a user query is received, the system embeds the query, searches the vector database for relevant documents, and retrieves the most relevant contexts. These contexts are then used to augment the original query, creating an enriched prompt that is passed to a large language model to generate a final response.

The key architectural components of a RAG system are

- *Document indexing*—We need to populate the knowledge base by indexing. Indexing transforms raw text into a machine-readable format by converting documents into high-dimensional vector embeddings. This vectorization process captures the semantic elements of the text and makes it easier to search using semantic queries.
- *Retrieval mechanism*—The "R" in RAG is the retrieval mechanism, which employs advanced vector similarity techniques to identify and rank the most relevant documents for a given query.